

TECHBIG SOLUTIONS

Hardware & Networking – Detailed Syllabus

Course Level: Beginner → Intermediate → Advanced

Recommended Duration: 120–180 Hours

Training Method: Theory + Practical Labs + Troubleshooting + Projects + Interview Preparation

Course Objective

A comprehensive, job-oriented Hardware and Networking program covering computer hardware, operating systems, network administration, IP addressing, switching, routing, wireless, security, servers, virtualization, cloud fundamentals, troubleshooting and practical projects.

MODULE 1 – Computer Fundamentals

- History and generations of computers
- Types of computers and computer architecture
- CPU, ALU, Control Unit, registers and buses
- Motherboard and system components
- Input, output and storage devices
- BIOS/UEFI and POST
- Ports and connectors: USB, HDMI, DisplayPort, VGA, audio and RJ-45

MODULE 2 – Computer Hardware Components

- Motherboard form factors and components
- CPU types, sockets and cooling
- RAM: DDR generations, DIMM/SODIMM, channels and troubleshooting
- Storage: HDD, SSD, NVMe and external drives
- Power supply units and power ratings
- Graphics cards and integrated graphics
- Expansion cards and peripheral devices

MODULE 3 – PC Assembly & Disassembly

- Safety and ESD precautions
- Identifying components and compatibility
- PC assembly step-by-step
- Cable management and front-panel connections
- BIOS/UEFI configuration
- POST testing and initial hardware validation
- Disassembly, cleaning and preventive maintenance

MODULE 4 – Operating Systems

- Operating system concepts
- Windows installation and configuration

- Windows editions and system requirements
- Disk partitioning and formatting
- Drivers and device management
- Windows utilities and administration
- Introduction to Linux and Linux distributions

MODULE 5 – Windows Administration

- User and group management
- File and folder permissions
- Task Manager and performance monitoring
- Services and startup management
- Event Viewer
- Device Manager
- Disk Management
- System Restore, recovery options and backup
- Windows command-line tools

MODULE 6 – Linux Fundamentals

- Linux architecture and distributions
- Installation and dual-boot concepts
- Linux file system
- Common commands and shell basics
- Users, groups and permissions
- Package management
- Processes and services
- SSH basics and basic troubleshooting

MODULE 7 – Storage & Data Management

- Storage technologies and interfaces
- Partitioning: MBR and GPT
- File systems: NTFS, FAT32, exFAT and Linux file systems
- Disk health and SMART concepts
- Disk cloning and imaging
- Backup strategies
- RAID concepts and levels
- Data recovery fundamentals

MODULE 8 – Hardware Troubleshooting

- Troubleshooting methodology
- No power and no POST problems
- Boot failures and blue-screen issues
- RAM, CPU and motherboard diagnosis

- Hard disk/SSD failures
- Display and graphics problems
- Keyboard, mouse, printer and USB troubleshooting
- Overheating, fan and power problems
- Diagnostic tools and logs

MODULE 9 – Networking Fundamentals

- Network concepts and terminology
- LAN, WAN, MAN, PAN and WLAN
- Client-server and peer-to-peer networks
- Network topologies
- Network devices and their roles
- Bandwidth, latency, throughput and packet loss

MODULE 10 – OSI & TCP/IP Models

- OSI seven-layer model
- TCP/IP model
- Functions of each layer
- Encapsulation and decapsulation
- Frames, packets, segments and bits
- Protocols mapped to network layers

MODULE 11 – Network Media & Cabling

- Twisted-pair cables: Cat5e, Cat6 and Cat6A
- Fiber-optic cables
- Coaxial cable concepts
- RJ-45 connectors
- T568A/T568B standards
- Straight-through and crossover concepts
- Cable testing and fault identification
- Structured cabling basics

MODULE 12 – Ethernet & Switching

- Ethernet fundamentals
- MAC addresses
- Switches and switching concepts
- Collision and broadcast domains
- ARP basics
- VLAN concepts
- Access and trunk ports
- STP concepts
- Basic managed-switch configuration

MODULE 13 – IP Addressing

- IPv4 addressing
- Public and private IP addresses
- Subnet masks
- CIDR notation
- Default gateway
- Network and broadcast addresses
- Subnetting and VLSM basics
- IPv6 fundamentals and address types

MODULE 14 – Network Protocols

- TCP and UDP
- HTTP and HTTPS
- DNS
- DHCP
- FTP/SFTP concepts
- SMTP, POP3 and IMAP concepts
- SSH and Telnet
- ICMP
- ARP and NTP fundamentals

MODULE 15 – Routing

- Routing fundamentals
- Static and dynamic routing concepts
- Default routes
- Routing tables
- Inter-network communication
- Introduction to RIP, OSPF and BGP
- Basic router configuration and troubleshooting

MODULE 16 – Wireless Networking

- Wi-Fi standards and generations
- 2.4 GHz and 5 GHz bands
- Access points and wireless controllers
- SSID and wireless security
- WPA2/WPA3 concepts
- Wireless channel planning
- Signal strength and interference
- Wi-Fi troubleshooting

MODULE 17 – Network Security

- Security fundamentals
- Threats, vulnerabilities and risks
- Firewalls
- Antivirus and endpoint protection
- Authentication and authorization
- MFA concepts
- VPN concepts
- Encryption basics
- Secure passwords and security best practices

MODULE 18 – Network Services & Administration

- DHCP server concepts
- DNS server concepts
- Web server basics
- File and print services
- Directory services and Active Directory fundamentals
- Group Policy concepts
- User and computer administration
- Network resource sharing

MODULE 19 – Active Directory

- Domain, forest and organizational unit concepts
- Domain controllers
- User, group and computer accounts
- Security groups and distribution groups
- Group Policy Objects
- Joining computers to a domain
- Basic Active Directory troubleshooting

MODULE 20 – Virtualization

- Virtualization concepts
- Hypervisors
- Virtual machines and virtual networks
- VMware/Hyper-V concepts
- Snapshots and virtual disks
- Resource allocation
- Virtualization troubleshooting

MODULE 21 – Cloud & Modern Infrastructure Basics

- Cloud computing concepts
- IaaS, PaaS and SaaS
- Public, private and hybrid cloud

- Virtual networks and cloud security basics
- Cloud storage concepts
- Introduction to AWS, Microsoft Azure and Google Cloud
- Hybrid infrastructure concepts

MODULE 22 – Network Monitoring & Troubleshooting

- Ping, tracert/traceroute, ipconfig/ifconfig and nslookup
- netstat and arp tools
- DNS troubleshooting
- IP configuration troubleshooting
- Connectivity and routing diagnosis
- Packet capture concepts
- Network monitoring and performance metrics
- Introduction to Wireshark

MODULE 23 – Server Fundamentals

- Server hardware components
- Rack and tower servers
- RAID and server storage
- Server power and cooling
- Windows Server fundamentals
- Linux server fundamentals
- Server roles and services
- Basic server maintenance

MODULE 24 – Backup, Recovery & Disaster Recovery

- Backup types: full, incremental and differential
- Backup schedules
- Local and cloud backup concepts
- System imaging
- Recovery procedures
- Business continuity basics
- Disaster recovery planning and RPO/RTO concepts

MODULE 25 – Printer & Peripheral Support

- Printer types and technologies
- Printer installation and drivers
- Network printer configuration
- Print queues and spooler
- Scanner and multifunction devices
- USB and Bluetooth peripherals
- Common peripheral troubleshooting

MODULE 26 – Preventive Maintenance

- Hardware cleaning procedures
- Thermal management
- Firmware and driver updates
- Disk health monitoring
- Power protection: UPS and surge protection
- Maintenance schedules
- Asset labeling and inventory management

MODULE 27 – IT Support & Help Desk

- IT support fundamentals
- Ticketing systems and incident management
- Incident, problem and change concepts
- Remote support
- User communication and escalation
- Documentation and knowledge bases
- Introduction to ITIL practices

MODULE 28 – Practical Labs

- PC assembly and OS installation
- RAM/storage replacement
- BIOS/UEFI configuration
- Windows and Linux administration
- IP addressing and subnetting exercises
- Switch and router configuration
- Wi-Fi setup and troubleshooting
- File/printer sharing
- Network troubleshooting using command-line tools
- Basic Wireshark packet analysis

MODULE 29 – Industry Projects

- Design and configure a small office LAN
- Build a Windows/Linux workstation
- Set up a home/office Wi-Fi network
- Implement a basic server and shared storage
- Configure a small business network with VLANs
- Create a backup and recovery plan
- Troubleshoot a simulated enterprise network

MODULE 30 – Interview & Certification Preparation

- Hardware and troubleshooting interview questions

- Networking fundamentals and scenario questions
- OSI/TCP-IP interview preparation
- IP addressing and subnetting tests
- Windows/Linux administration questions
- Network security fundamentals
- Practical troubleshooting assessments
- Resume and technical interview preparation

CAREER & TRAINING OUTCOMES

- Computer Hardware Technician
- Desktop Support / IT Support Engineer
- Network Support Engineer
- System Administrator
- Network Administrator
- Technical Support Engineer
- Junior Server Administrator
- IT Infrastructure Support

Students completing the program will be able to assemble and troubleshoot PCs, install and administer operating systems, configure LAN/Wi-Fi networks, perform IP addressing and subnetting, troubleshoot network connectivity, understand routing and switching, apply security practices, support servers and virtualization environments, and complete practical infrastructure projects.

TECHBIG SOLUTIONS

Professional Training • Practical Learning • Industry-Oriented Skills

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